BROWN AIR CHARGE SYSTEM (BACS)

EXECUTIVE SUMMARY

The ***BROWN AIR CHARGE SYSTEM (BACS)*** installed in the air induction system of an internal combustion engine, provides measurable, and often, dramatic combustion improvements.

These significant combustion improvements are manifested by increases in ***torque, horsepower, and brake specific fuel consumption (bfsc).*** The BACS induced combustion improvements simultaneously ***lead to significant reductions in environmentally undesirable noxious emissions***, including: ***HC, CO, NOx, CO_2, SO_2, particulates, visible smoke and aldehydes.***

The BACS is installed in the intake air stream so that the air entering the combustion chamber is provided with ***combustion stimulating molecules and radicals*** ***(CSMR)***, including ***OH radicals***, ***Hydrogen Peroxide (H_2O_2), Oxygen (O_2) as Oxygen Enriched Air (OEA), and Ozone (O_3)*** molecules. ***These molecules provide powerful combustion stimulation***, hence ***CSMR***. The BACS generated ***OEA*** and ***CSMR*** aspects also increase the reduction efficiency of converter equipped vehicles re ***HC*** and ***CO*** by a major to near 100% factor, without a separate oxygen supply system.

The inventor, [redacted] has accomplished the most definitive work, both in the field guided by [redacted] San Diego State University, and the laboratory. The laboratory phases have been under the general guidance of [redacted] Professional International Consulting Engineer, [redacted], retired Senior Research Scientist in Chemistry, NOAA, and other scientists. Laboratories at ***Southwest Research Institute***, San Antonio, TX; Jensen Beach Campus of ***Florida Institute Of Technology***; ***PetroSystems International***, Nashville, TN; the ***Mobile Sources Emissions Research Laboratory, EPA***, Research Triangle Park, NC; ***San Diego State University***, San Diego, CA; ***Transworld Carburetors, Inc.***, Taipei, Taiwan; the ***NITTSU SYOJI Co.***, Ltd. testing facility, Osaka, Japan, and a variety of independent laboratories have been utilized in the endeavor to fully understand and define the chemical and physical dynamics inherent in the BACS functions.

In addition to extensive laboratory confirmation, the documented results of actual field use of the BACS on both gasoline and diesel fueled engines, during several hundred marine engine hours, and several million-highway miles, are exceptionally positive. Based on the results of early research the basic objectives to be achieved by use of the BACS on heat engines were established as:

1) A minimum reduction of 10% in fuel consumption; plus
2) A minimum gain of 5% in effective shaft or road horsepower; plus
3) A minimum reduction of 25% in all controlled noxious emissions, including particulate matter, except for NOx (where the reductions were normally somewhat less than 25%).

Laboratory, stationary, marine, and highway use, established that the stated objectives were not only realistic minimums, but were exceeded in well over 90% of all monitored applications of the BACS. Typical are the results achieved during an extended marine research/demonstration project of the BACS on a medium size (2108 cid) turbo-charged, inter-cooled marine diesel engine. Results reflected that the BACS provided either a 20% reduction in fuel consumption, or a 20% gain in speed (achieved by increasing shaft horsepower) while simultaneously reducing diesel smoke and particulates by 40 plus percent under either option. The project results were monitored by an agency of the U. S. Government.

While the potential applications of the BACS are extremely broad, ***the most practical immediate uses***, in addition to ***trucks and buses***, lie with ***large, constant speed engines such as those used by marine vessels, generators, locomotives, mining equipment, and pumps used in flood control and irrigation. Future potential uses include the total spectrum of automotive applications, especially hybrid powered vehicles, ocean going vessels***, various ***military uses***, and ***of greatest importance***, ***large scale power generation plants and industrial furnaces.***

NOTE: **This summary reflects the basic results of twenty-five years of BACS combustion research. The OH radical generation aspect needs further research/confirmation.**

********** 2-10-99 **********

BACS Vs Federal Test Procedures

(FTP)

After a few years of extensive BACS research, which included several million miles of highway testing and several FTP prescribed laboratory tests, it was conclusively proven that the two procedures were basically not compatible. This contrast became especially evident when the BACS was tested on carburetor equipped gasoline engine vehicles under Federal Test Procedures.

At some point in the early research years it was also determined *WHY* the Power Pak element *DID NOT RESPOND* to FTP laboratory type testing. The common factors were that each of the test vehicles had gasoline fueled, carburetor equipped engines, and each FTP test included the overnight "Cold Soak", prior to initiation of the test series. The test results were uniformly of such a nature that the *BACS appeared as a device that could possibly enhance combustion efficiency, however, on a very small scale*. In contrast the *controlled actual highway tests*, which had included use of a variety of calibrated fuel economy measuring devices, *without exception, indicated BACS induced fuel economies far greater than those shown by the FTP results.* It was unknown in the early tests that upon starting the engines in the laboratory after the "Cold Soak" period, the engine intake air *was NOT routed through the Power Pak element* and therefore the air could NOT possibly be treated by the BACS unit. It was also determined that this situation prevailed until the engine warmed up and the heat riser control mechanism altered the air feed gate to allow ambient air into the engine. In essence the FTP "Cold Soak" mechanically negated the possibility of the Power Pak unit providing treated charge air for an extended period.

This major problem was further compounded in that it was later discovered that there was also a delay of from eight minutes upward in the Power Pak becoming fully functional and the *catalytic action* of the BACS "Thunderstorm In A Bottle" on the charge air becoming evident and measurable. The effect of this anomaly was later ameliorated somewhat when both the Scientific Advisor and the EPA Administrator recommended diesel research and testing as a more needed and possibly productive field. Their advice was followed.

BACS Vs Federal Test Procedures (FTP)

After several years of extensive BACS research, which included several million miles of highway testing and several FTP prescribed laboratory tests, it was conclusively proven that the two procedures were basically not compatible. This contrast became especially evident when the BACS was tested on carburetor equipped gasoline engine vehicles under Federal Test Procedures.

At some point in the first few years of BACS research it was determined why the Power Pak element DID NOT respond to FTP type testing in EPA approved commercial laboratories. Each of the test vehicles had gasoline fueled carburetor equipped engines, with each test including the overnight "Cold Soak", prior to initiation of the test series. The test results were uniformly of such a nature that the BACS appeared as a device that could possibly enhance combustion efficiency, however, on a very small scale. The controlled highway tests, which had included use of a variety of calibrated fuel economy measuring devices, without exception, indicated BACS induced fuel economies far greater than those shown by the FTP results. It was unknown at that time that upon starting those engines in the laboratory after the "Cold Soak" period, the <u>engine intake air</u> was NOT routed through the Power Pak element and therefore <u>could NOT possibly be treated by the BACS unit</u>. It was also determined that this situation prevailed until the engine warmed up and the heat riser control mechanism altered the air feed gate to allow ambient air into the engine. <u>In essence the FTP "Cold Soak" totally negated the possibility of the Power Pak unit providing treated charge air for an extended period</u>. This problem was compounded in that it was later discovered that there was a delay of from eight minutes upward in the Power Pak becoming fully functional and the catalytic action of the BACS " <u>Thunderstorm In A Bottle</u>" on the charge air becoming evident and measurable. The effect of this anomaly was ameliorated somewhat when the Scientific Advisor and the EPA Administrator both recommended diesel research and testing as a more needed and possibly productive field. Their advice was followed.

The very limited FTP results shown reflect the early successes while using the BACS under other than "Cold Soak" FTP type testing on gasoline and the first laboratory FTP diesel tests. The limited gasoline and diesel FTP tests are set forth with the older gasoline test results shown first

<u>FEDERAL TEST PROCEDURES</u>
<u>Gasoline</u>

M. E. R. C. TEST REPORT, 10-24-80-G is set forth in toto on the following three pages. The report covers both *<u>ACTUAL Highway Fuel Economy Road Tests</u>* as well as *<u>Highway Fuel Economy Tests (HFET) and Steady State Tests (SS) under Federal Test Procedures,</u>* in an EPA approved laboratory.

MOBILE ENERGY RESEARCH CENTER

power pak

AIR CHARGE SYSTEM® PRE-COOLER

[redacted]

M.E.R.C TEST REPORT
10-24-80-G
HIGHWAY FUEL ECONOMY TESTS (Road Tests)
and
HIGHWAY FUEL ECONOMY TESTS (Lab)
Federal Test Procedures

1) ***TEST VEHICLE:*** 1979 Dodge Colt, equipped with a 98 CID, gasoline engine, and 8 speed manual transmission. VIN: 4H24K95300992.

2) ***PURPOSE OF TESTS:*** Pure research, to determine the possible effect of installation of the Power Pak Air Charge System (PPACS) on the fuel economy, emissions, and driveability of test vehicle under actual highway operating conditions, and the simulation of highway operating conditions by Federal Test Procedures in a laboratory.

3) ***CHRONOLOGY:*** ***ACTUAL HIGHWAY FUEL ECONOMY***

In early 1979 two changes were made to the stock 1979 Dodge Colt: Alaskan Brand Racing Oil and Coolant were installed, followed by installation of a Jacobs Compusensor (computerized electronic ignition system), along with Jacobs ignition wire and a Jacobs coil. Extensive records of the test vehicle, after the two changes, and prior to installation of the Power Pak, confirmed city mileage of 28-30 MPG and highway mileage of 38-42 MPG.

BASELINE VERSUS POWER PAK RUNS

On Aug. 7, 1979, after a baseline run, with results as shown below, a Power Pak unit was installed and a series of three consecutive test runs were completed. The results are as shown:

	Miles	%City	Miles	%HWY	Tot Miles	Avg. MPG
Baseline	45	53%	40	47%	85	35.54
* Run #2	22	28.5	55	71.5	77	48.00
* Run #3	29.2	75	10	25	39.2	52.26
* Run #4	1.7	5	32.2	95	33.9	61.64
					235.1	

* With Power Pak installed.

Fuel consumption during the above tests was measured by filling the tank to an overflow condition on refill after each run, and then waiting appx. 5 minutes to be sure the fuel could be touched by a finger. The dollar total on the pump was then divided by the price per gallon to obtain the precise volume used.

The above tests were conducted by Charles L. Brown, President, CBE, Inc.. The runs were observed and verified by Mr. Daniel W. Christoffersen, President, Alaskan Brand Oil Sales. All city speeds were observed as posted; the highway speeds averaged over 55 mph.

After these runs the Power Pak was removed from the vehicle until completion of the baseline FTP Tests, conducted at AESI on Aug. 27, at which point the unit was re-installed. The Power Pak remained on the test vehicle from 9,300 miles to 39,000 miles; during which period an extensive number of highway fuel economy tests were conducted. The vehicle was subsequently officially clocked at an average speed, during two direction runs, of ***106 MPH*** at the Dry Lakes/Mirage, California Race Facility.

4) ***TEST PROCEDURES:*** Utilized the services of AUTOMOTIVE ENVIRONMENTAL SYSTEMS, INC. (AESI) of Westminster, CA, a laboratory on the EPA list of recognized test facilities, to conduct Highway Fuel Economy Tests (HFET) and Steady State Tests (SS) (PPACS Only) on test vehicle, both prior to and after installation of the PPACS unit. Testing procedures were those prescribed by the Federal Test Procedures and are as described in the Federal Register, Part 600, Subparagraph G.

FTP BASELINE TESTS: Aug. 27, 1979.
(9,300 miles)

	HC	CO	CO_2	NOxC	MPG
Test 1:	.505	11.844	208.106	1.655	38.840
Test 2:	.586	14.800	202.632	1.662	38.938
Avg. :	.549	13.322	205.369	1.658	38.889

POWER PAK AIR CHARGE SYSTEM TESTS: May 12, 1980
(39,000 miles)

	HC	CO	CO_2	NOxC	MPG
	.526	3.617	N/A	2.141	44.039

Net change due to PPACS use:

	HC	CO	CO_2	NOxC	MPG
	-3.5%	-76%	N/A	+ 29%	+ 13%

TECHNICAL NOTE: The 1979 Dodge Colt used as a test vehicle is equipped with a unique 8 speed gearing arrangement. The first 4 speeds are in the Power Mode, with the second 4 speeds classified as the Economy Mode. It is ***unfortunate*** that the two baseline tests on Aug.27, 1979 were conducted in the ***ECONOMY MODE,*** whereas the tests of

May 12, 1980 with PPACS were conducted in the ***POWER MODE***. The higher rpm's of the Power Mode are believed responsible for the higher NOxC with the Power Pak installed; a unique, one of a kind, incident experienced during extensive testing of the concept.

COMPLETE POWER PAK AESI TEST SERIES
Federal Test Procedures

May 12, 1980

Test	HC	CO	NOxC	MPG	Comments
HFET	0.526	3.617	2.141	44.039	1.25" venturi device
40 MPH Steady State	0.368	1.757	2.716	52.061	1.25" venturi device
55 MPH Steady State	0.406	0.888	2.025	43.455	1.25" venturi device
65 MPH Steady State	0.324	3.748	3.402	39.130	1.25" venturi device
55 MPH Steady State	0.415	1.622	2.037	43.770	1.00" venturi device

SUPPORTING DOCUMENTATION: Copies of the AESI, Inc. reports, and other documentation as set forth herein, are maintained in the CBE/MERC Corporate files in Miami, Florida.

The above report is certified accurate to the best of my knowledge and belief at Miami, Dade County, Florida on this the 24th day of October, 1980.

[redacted]
President, C.B.E., Inc.
Director, M.E.R.C.

EPA EMISSIONS RESEARCH LABORATORY

Diesel Federal Test Procedures (FTP)

Perhaps the most significant controlled test data available, relative to the effects of the Brown PPACS Pre-Cooler on diesel engines, is set forth in the following tables and notes. The data in the tables was accumulated between Oct. 20—22, 1982 and Aug. 23—Sept. 1, 1983 at the Environmental Research Center, Mobile Sources Emissions Research Laboratory, Environmental Protection Agency, Research Triangle Park, North Carolina. Although FEDERAL TEST PROCEDURES (FTP) were utilized, the primary purpose of the test series was of a purely scientific nature to determine if the Brown PPACS Pre-Cooler, aka BROWN AIR CHARGE SYSTEM (BACS), could reduce diesel gaseous and particulate emissions. The *positive nature of the results are believed to be a scientific first for simultaneous reductions of all diesel emissions, and for the exceptionally positive effects that the air treatment process had on improving fuel economy in essentially all driving modes.* Full documentation of the test results is maintained in the MERC/CBE, Inc. files.

SELECTED RESULTS OF: BROWN POWER PAK AIR CHARGE SYSTEM®PRE-COOLER

BACS (Combustion Catalyst)

DIESEL EMISSIONS RESEARCH TESTS

CONDUCTED BY: Environmental Research Center, Mobile Sources Emissions Research Laboratory, EPA, Research Triangle Park, North Carolina – Oct. 20-22, 1982 and Aug. 23 – Sept.1, 1983.

TEST VEHICLE: 1981 Chevrolet Chevette (1.8 Liter Isuzu Diesel)

NEW YORK CITY CYCLE (1982)

	HC	CO	NOx	C02	PART.	MPG
			Grams per vehicle mile (GPM)			
BASELINE	.173	1.75	2.56	461.67	.6888	21.88
PPACS +MFC	.119	1.44	2.40	422.43	.645	23.93
NET CHANGE	**-31.2%**	**-17.7%**	**-6.2%**	**-8.5%**	**-6.2%**	**+9.4%**

LOS ANGELES CYCLE (1983)

	HC	CO	NOX	CO2	PART.	MPG
BASELINE	.344	1.590	1.544	341.165	N/A	29.51
PPACS	.215	.943	1.469	288.749	N/A	34.97
NET CHANGE	-37.5%	-40.7%	-5.5%	-15.36.%	N/A	+18.5%

LOS ANGELES –4 (STABALIZED TEST #2) (1983)

	HC	CO	NOX	CO2	PART.	MPG
BASELINE	.346	1.086	1.748	346.087	N/A	29.16
PPACS	.218	.787	1.553	280.434	N/A	36.03
NET CHANGE	-37%	-27.5%	-11.2%	-18.97%	N/A	+23.6%

POWER PAK (BACS)

EPA LAB TEST SUMMARY

	HC	CO	NOX	CO2	PART.	MPG
NYCC	-31.2%	-17.7%	-6.2%	-8.5%	-6/2%	+9.4%
LA-4:	-37.5%	-40.7%	-5.5%	-15.36%	N/A	+18.5%
LA-4 (ST-2)	-37%	-27.5%	-11.2%	-18.9%	N/A	+23.6%

New York City Cycle (NYCC): Distance 1.2 Miles: Time 10 Minutes; Avg. Speed 7.1 MPH; Stops Per Mile 9.32; Idle Time 40%
LA-4: Distance: 7.5 Miles; Time 22.9 Min.; Stops Per Mile 2.4; Avg. Speed 19.7 MPH
LA-4 (ST-2); Distance 3.9 Miles: Time 14.5 Min.; Stops Per Mile 3.08; Avg. Speed 16.1 MPH

****End of Limited EPA Test Results re PPACS/BACS Technology****